About Me

Hey! I'm a Computer Engineering student at the University of Texas at Arlington, graduating May 2026. I love building things that bridge the gap between code and the physical world.

Whether it's programming microcontrollers, working on autonomous robots, or diving into machine learning research, I'm happiest when I'm solving problems that actually **do something** in the real world.

Right now, I'm working on an autonomous ground robot for the IGVC competition and exploring AI/IoT security through research. When I'm not debugging firmware or tuning navigation algorithms, you'll find me competing in hackathons, building quirky apps and websites that solve my own problems (or are just plain fun), or practicing guitar — slowly learning to speak the language that comes straight from the heart.

What I Do

// Real-time embedded systems on ARM Cortex-M

// Autonomous robotics with ROS2 and sensor fusion

// Machine learning & neural network development

// Research in AI-enabled IoT and optimization systems

Technical Skills

Machine Learning

- Python, PyTorch
- Neural networks & model evaluation
- Multi-fidelity surrogate models
- Data analysis & visualization

Embedded & Firmware

- Bare-metal ARM (STM32, TM4C)
- FreeRTOS, interrupts, memory
- DSP & real-time audio processing
- Sensor integration & motor control

Languages

- C, C++
- Python, Bash
- ARM Assembly
- Version Control: Git/GitHub

Interfaces & Peripherals

- SPI, I2C, UART
- PWM, ADC/DAC, GPIO
- I2S audio streaming
- Sensor protocols (GPS, IMU, LiDAR)

Debugging & Tools

- Oscilloscope, Logic Analyzer
- GDB, CMake, Linux CLI
- Code Composer Studio
- Custom PCB bring-up & validation

Robotics & Systems

- ROS2, computer vision
- GPS/IMU/LiDAR fusion
- Navigation algorithms
- Real-time perception pipelines

Quick Stats

GRADUATING	FOCUS	LANGUAGES
May 2026	Embedded & Robotics	C/C++, Python

HARDWARE
STM32, TM4C, Jetson

RESEARCH
AI/IoT Security

My Philosophy

- Build to learn
- Debug with data
- Ship things that work
- Stay curious

I believe the best way to understand something is to build it. Whether it's a custom RTOS kernel or an autonomous robot, I'm driven by the challenge of making complex systems actually work.

Resume / CV

[PDF Format]

Research & Publications

Publications & Manuscripts

Submitted

Dec 2025

A Novel Optimization Model for Cost-effective Sustainable Power Plant

// First Author | Submitted

Developed a Direct Numerical Solution Optimization Algorithm (DNSOA) and compared performance against 7 Nature-Inspired Meta-Heuristic Multi-Objective Algorithms (e.g., NSGA-II, PSO). Integrated LSTM neural networks to predict future source availability, achieving a 0.12% reduction in system uncertainty and an ~88% reduction in pollutant emissions.

Submitted

Jan 2024

Investigation of Turbulence Model in CFD Flow for NACA 2412 Airfoil at High Reynolds Number

// First Author | Submitted

Conducted a CFD study on NACA 2412 airfoils at high Reynolds numbers (1.3×10^6) to analyze lift, drag, and aerodynamic efficiency. Evaluated 11 RANS turbulence models utilizing a Multi-Criteria Decision Making (MCDM) algorithm based on TOPSIS to identify optimal solvers.

Submitted

Jan 2024

Multi-Criteria Decision Making-Based Selection of RANS Turbulence Models for CFD Analysis of NACA 2412 Airfoil

// First Author | Submitted

Analyzed airflow at a moderate Reynolds number of 6.6×10^5 using hybrid mesh topology in ANSYS Fluent. Applied MCDM algorithms to rank 11 RANS models, determining the Transition SST model achieved the least deviation (9.23%) against X-Foil data.

Submitted

Dec 2025

The Secure AI-Enabled IoT Edge: Attacks, Lightweight Defenses, and Privacy Challenges

// Systematic Review | Submitted

Co-authored a systematic review on AI-enabled IoT edge security. Synthesized threats, lightweight defenses, and privacy-preserving learning practices across peer-reviewed literature.

Research Experience

Research Assistant

University of Texas at Arlington Research Institute (UTARI)

Jan 2023 – Mar 2023

- Developed multi-fidelity neural networks achieving >92% predictive accuracy
- Reduced reliance on high-cost simulations by 60% through model-driven surrogate predictions
- Collaborated in R&D workflow with iterative experimentation and reporting

UR2PhD Research Training

Sep 2025

Computing Research Association (CRA)

Completed UR2PhD Undergraduate Pre-Research Experience program focused on reading technical papers, conducting literature searches, and analyzing/visualizing experimental data.

Education & Recognition

UNIVERSITY	DEGREE	GRADUATION
UT Arlington	B.S. Computer Engineering	May 2026

HONORS

Honor's College,
Maverick Scholarship

Selected Projects

2025-Present Status: IN_PROGRESS

Autonomous Outdoor Robot (IGVC)

Goal

Model optimization, safety integration, & autonomy

Technologies

Jetson Orin NX, TM4C123, LiDAR, Computer Vision, C++/Python

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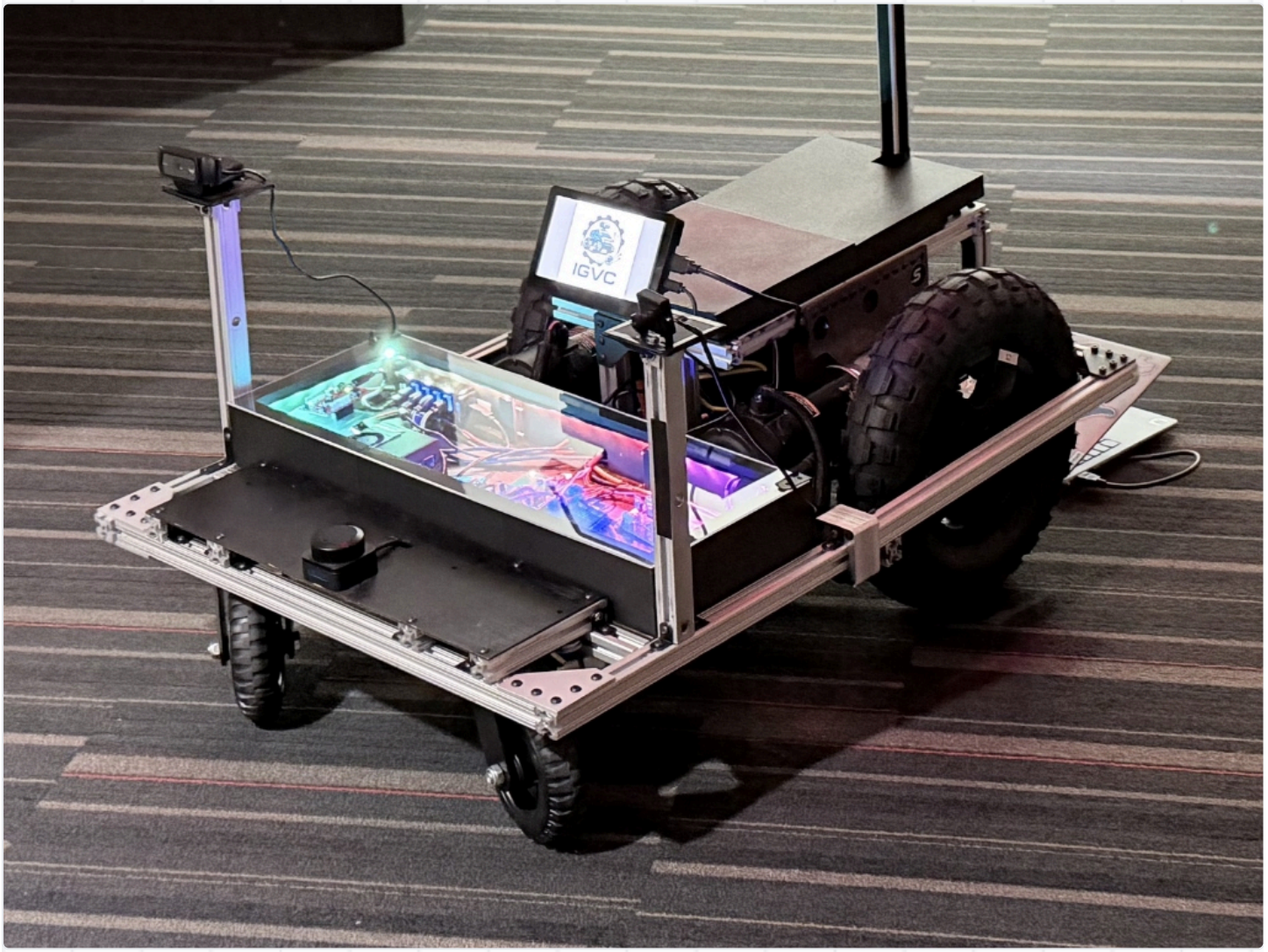

Developing a unified autonomous mode for the Intelligent Ground Vehicle Competition. The 3-layer architecture (Input, Processing, Output) integrates signals from LiDAR, cameras, and encoders for real-time decision making.

Focus: Enhancing safety systems with wireless E-stops and warning lights, while optimizing the lane detection and obstacle avoidance algorithms on the Jetson Orin

NX.

[See More] ↗

2025-12 Status: COMPLETED

Real-Time Operating System (RTOS)

Goal

Preemptive kernel with MPU & Shell on ARM Cortex-M4F

Technologies

C, ARM Assembly, TM4C123, Memory Protection

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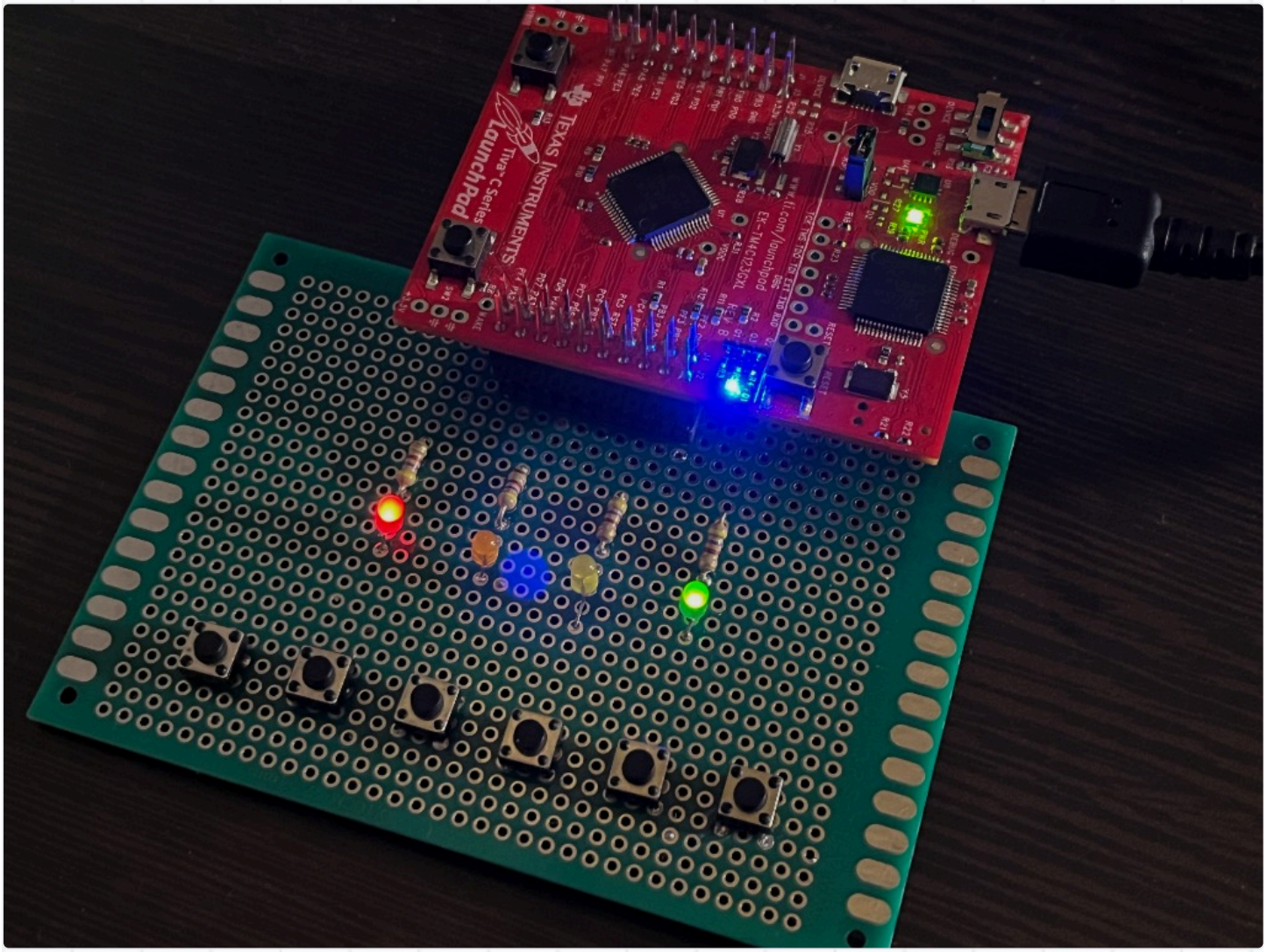


Developed a custom preemptive RTOS kernel from scratch for the TM4C123 (ARM Cortex-M4F). Features include an 8-level priority scheduler, inter-process communication (mutexes/semaphores), and hardware memory protection.

Highlights: Implemented a CLI shell for process management and a custom heap allocator with MPU-enforced access rights.

[See More] ↗

2025-11 Status: COMPLETED

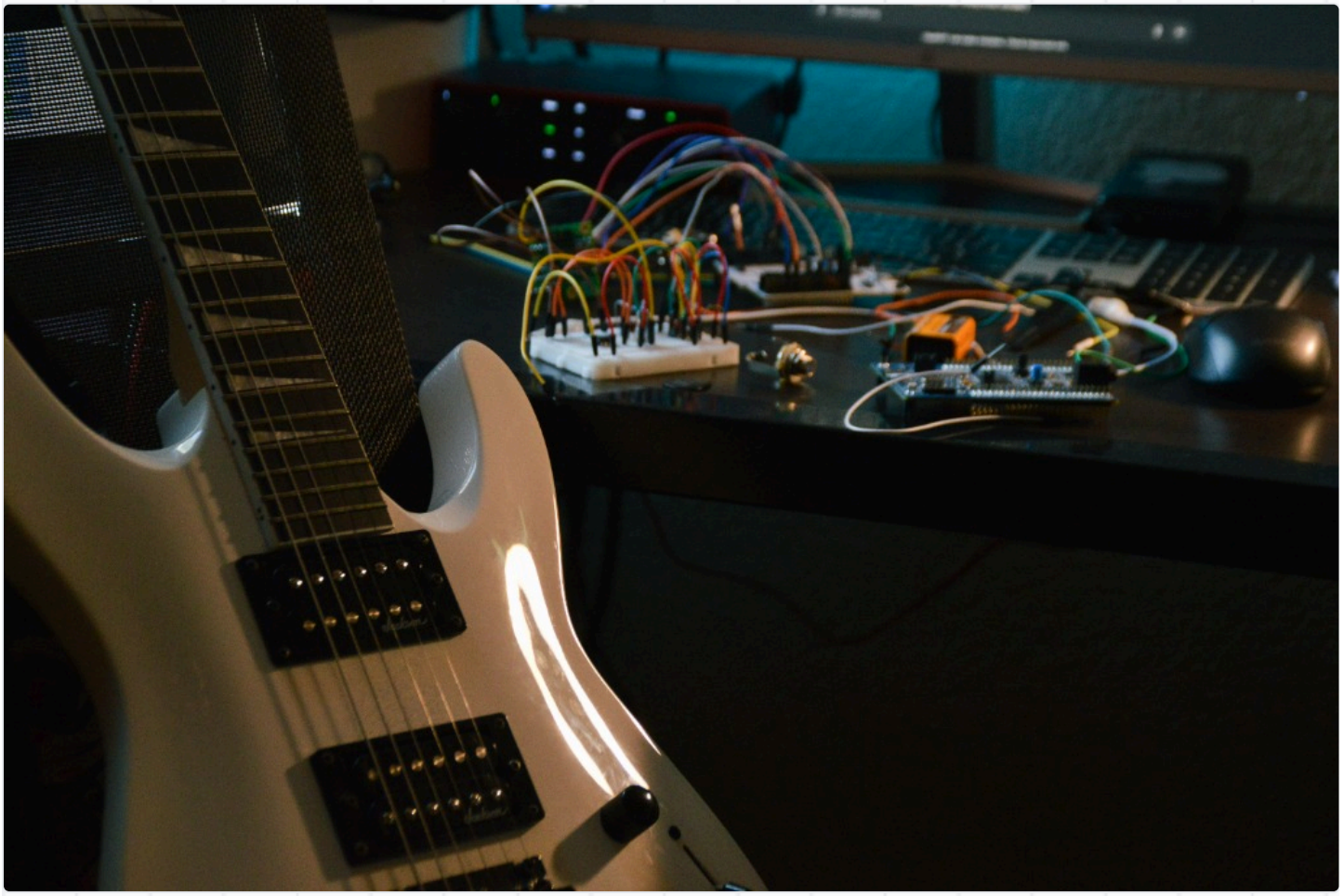
Hybrid Guitar Distortion Pedal

Goal

Analog signal path with STM32-based digital control

Technologies

STM32F446RE, TL082 Op-Amp, Relay Switching, Analog Circuits, C



Designed a hybrid effects pedal that combines a pure analog distortion circuit (TL082 op-amp + diode clipping) with an STM32 microcontroller for precise digital control of signal switching.

Philosophy: Preserved the "warm" organic tone of analog circuits while identifying that digital control (via relays) offers superior reliability for muting/bypassing without digitizing the audio signal.

[See More] ↗

2024-11 Status: COMPLETED

Embedded LCR Meter

Goal

Measure R, L, & C via comparator interrupts & precise timing

Technologies

C, TM4C123, Hardware Timers, Analog Comparators

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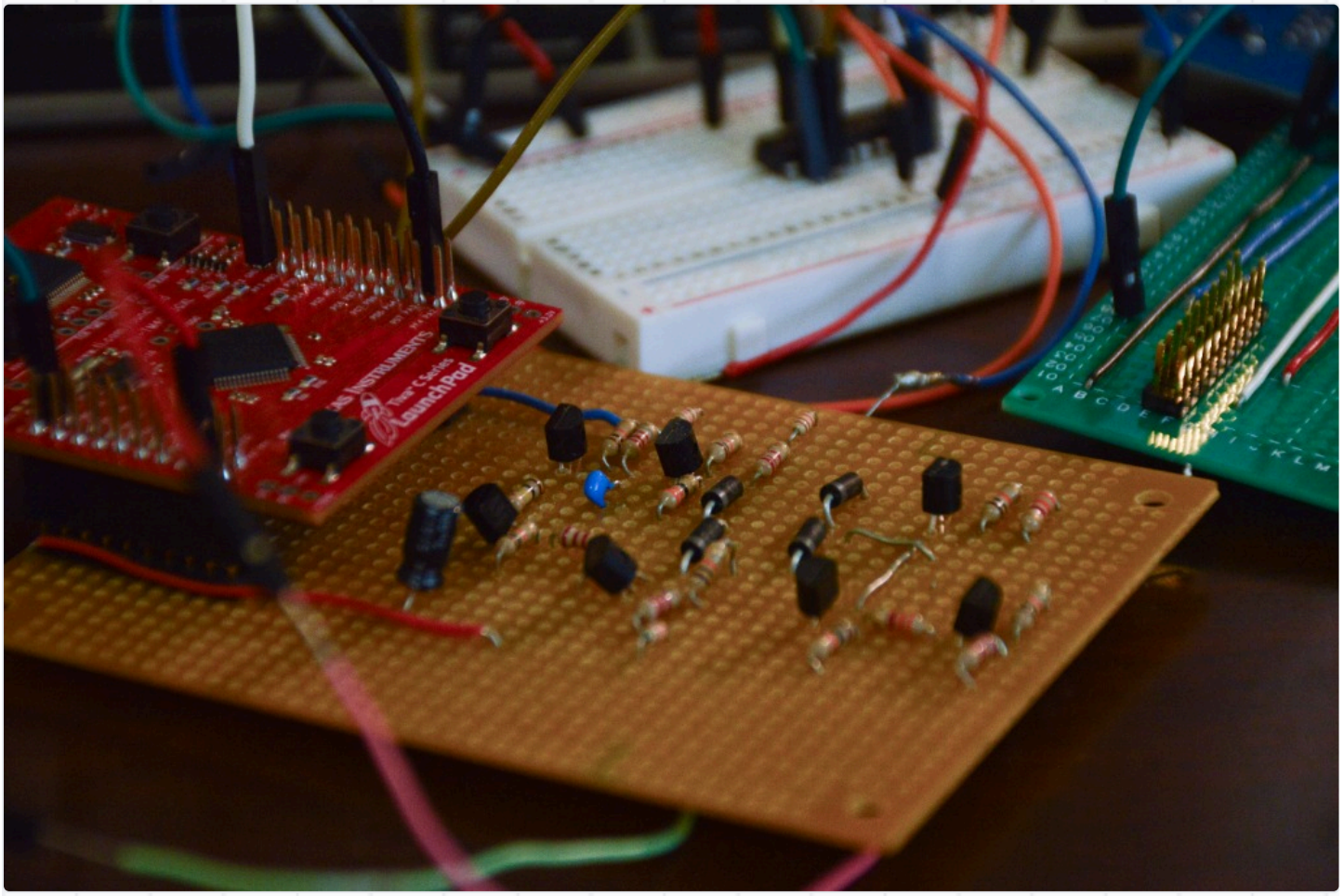


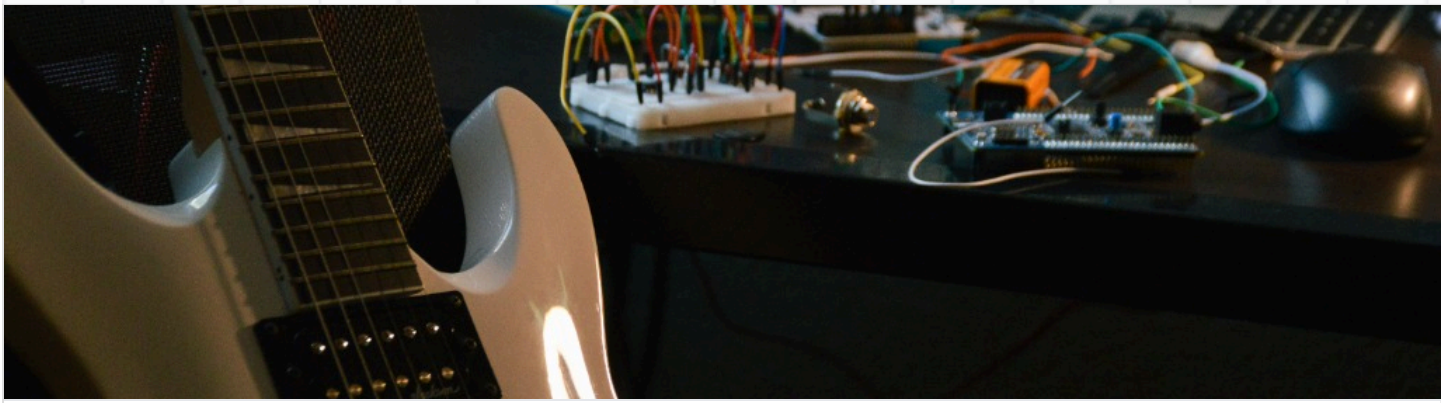
Implemented a precision LCR meter on an ARM Cortex-M4 microcontroller. Uses a text-based CLI to trigger measurement cycles for Resistance, Capacitance, and Inductance by timing charge/discharge curves.

Key Tech: Replaced traditional ADC sampling with **Analog Comparator Interrupts** to capture sub-microsecond timing events for calculating component values.

[See More] ↗

Silicon Arsenal

My daily drivers for bringing code to the physical world.

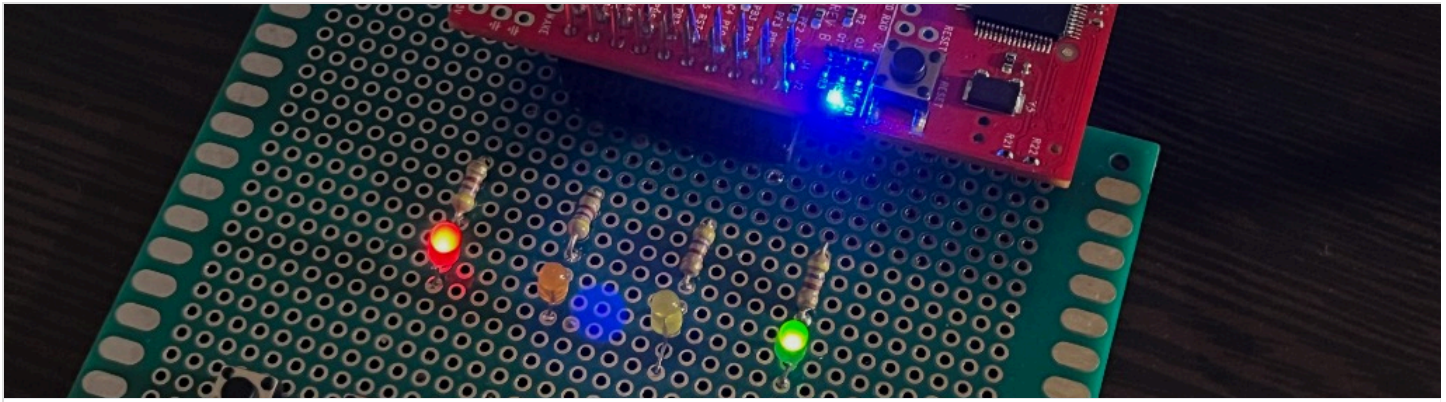


STM32 Nucleo-64

// The Workhorse

ARM Cortex-M4 @ 180MHz | FPU & DSP Instructions | 16-bit ADCs

Primary choice for audio processing and complex control loops requiring deterministic timing.

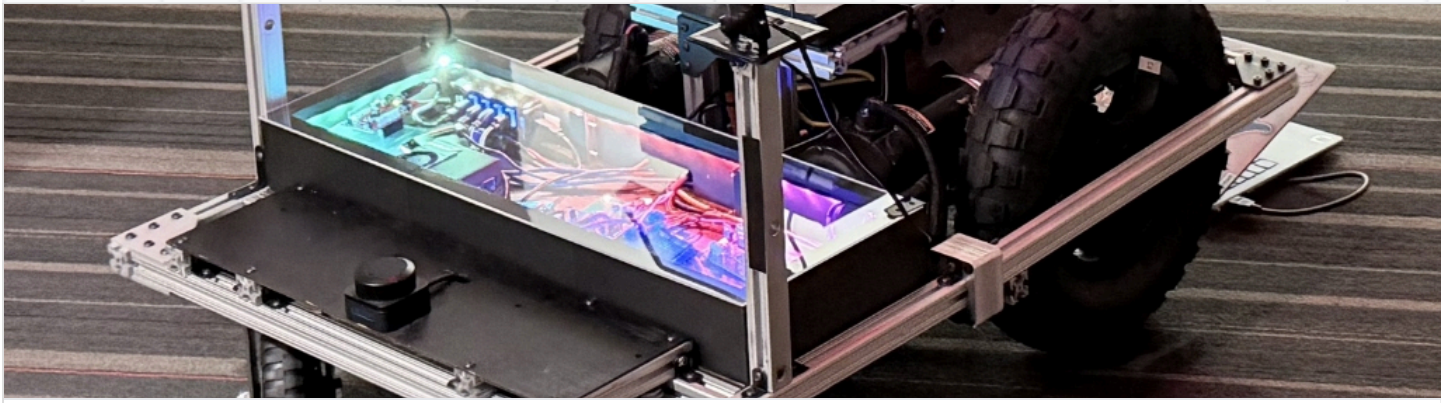


Tiva C Series (TM4C123)

// Real-Time Control

ARM Cortex-M4F @ 80MHz | Double Precision FPU | High-Drive GPIOs

Go-to platform for RTOS development and educational embedded systems projects.

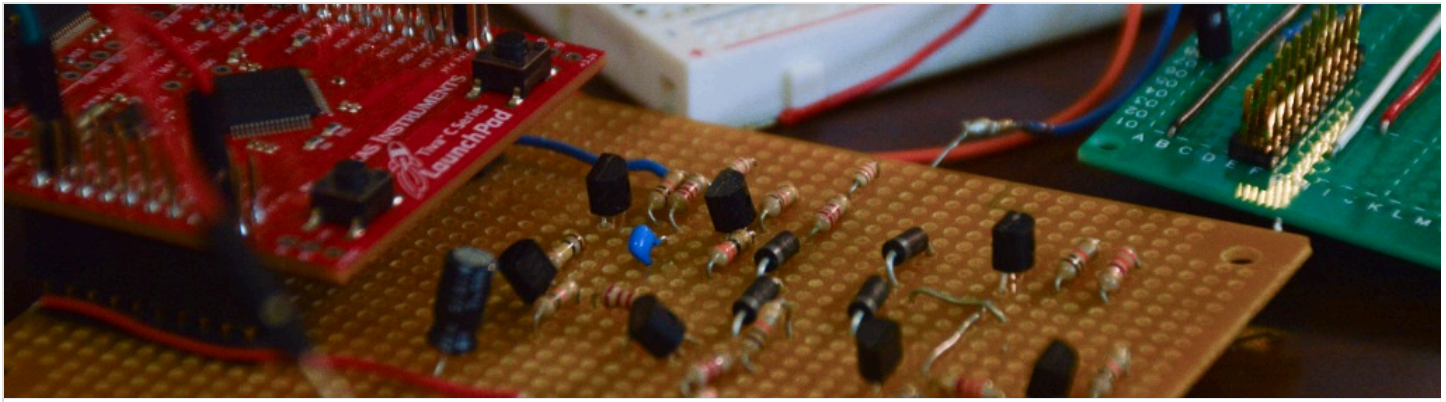


NVIDIA Jetson Orin NX

// AI at the Edge

100 TOPS AI Performance | Ampere GPU Architecture | ROS 2 Support

Sensor fusion and path planning for autonomous robotics (IGVC).



Cyclone IV FPGA

// Digital Logic Fabric

6K Logic Elements | Embedded Multipliers | PLLs for Clock Gen

Custom digital peripherals and high-speed parallel I/O interfaces.

Lab Instrumentation

[Tektronix Oscilloscope] [Logic Analyzer]

[Soldering Station] [Digital Multimeter]

Creative Lab

// Experiments in interface, AI, and design.

LEVEL 1

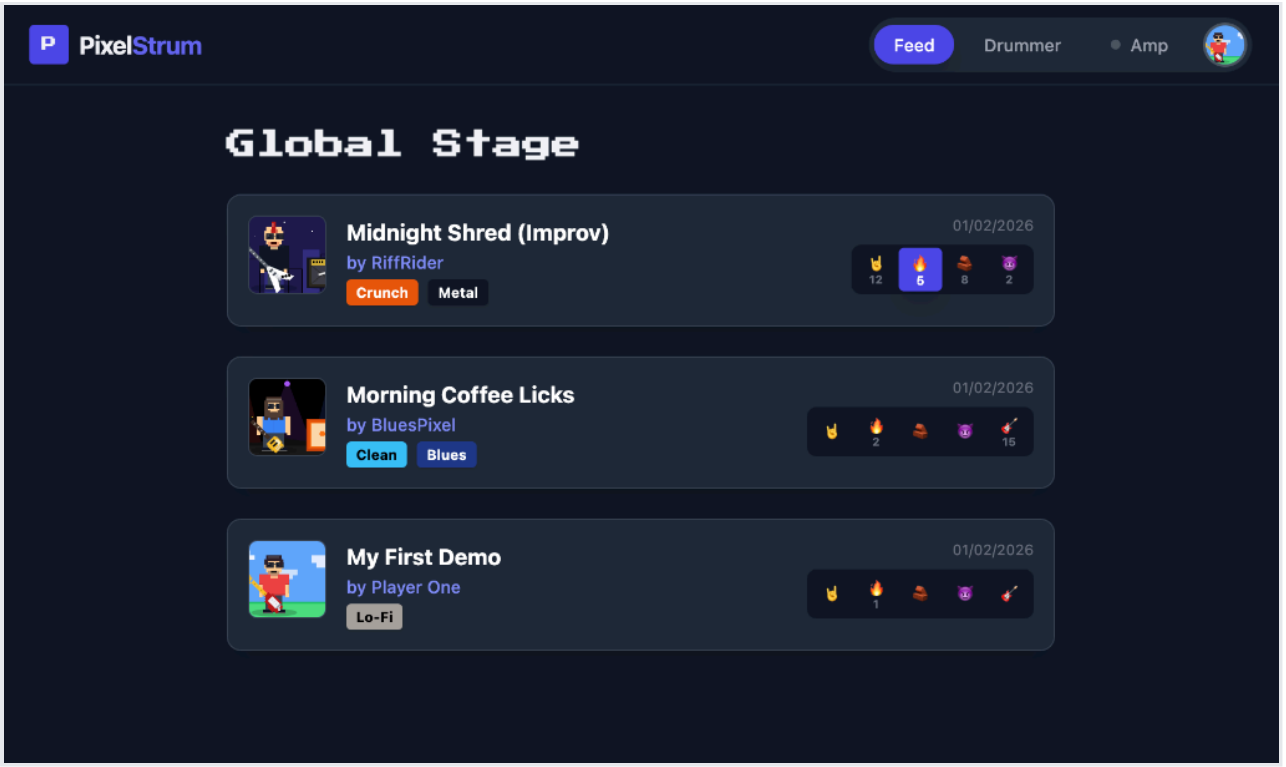
PixelStrum

QUEST Gamified social music platform with AI integration

PixelStrum is a concept web application that bridges the gap between social networking, gamification, and serious musical utility for guitarists.

Focus: A matched "retro-gaming" aesthetic meets advanced AI tools like an Avatar Creator, Session Drummer, and "NeuroShred" Amp Sim.

START GAME



Bruce

v2026-01

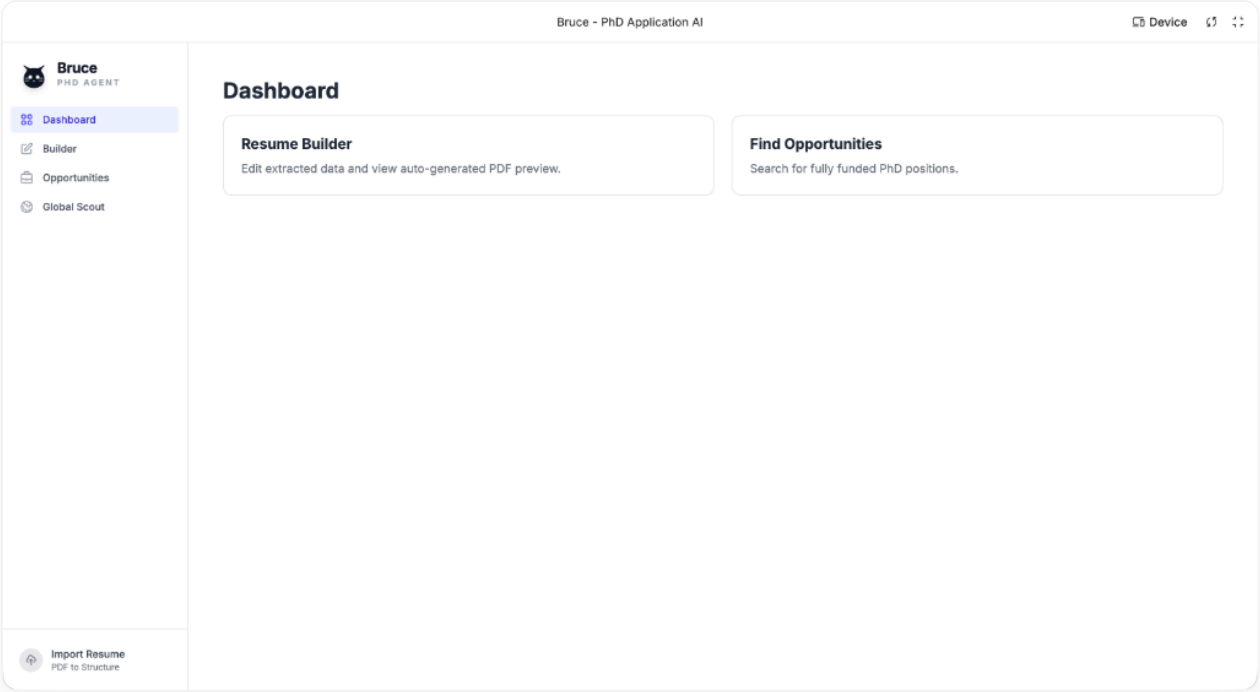
SYSTEM OBJECTIVE

AI-Powered PhD Application Consultant

An intelligent "PhD Application AI" designed to act as a personal wingman for aspiring doctoral students. Built on the cutting-edge Google Gemini 2.0 Flash API.

Mission: Streamline the chaotic journey from resume creation to finding the perfect supervisor and crafting the cold email.

Deploy Agent



SyncSpace

BETA RELEASE

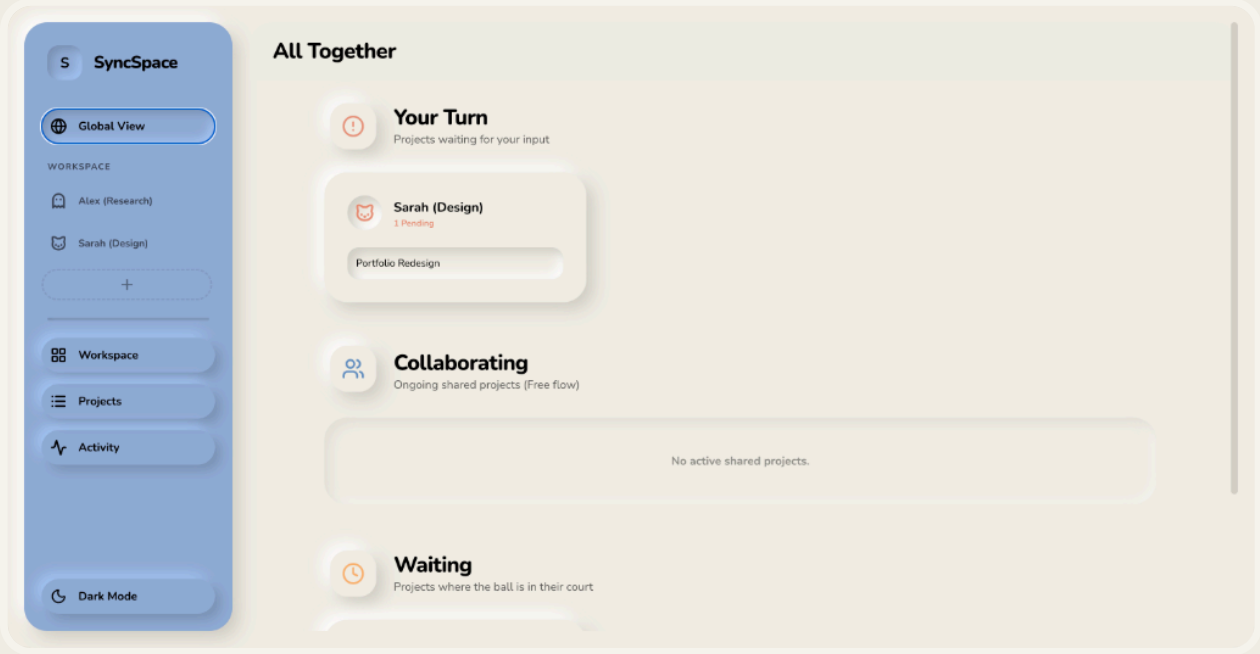
SYSTEM OBJECTIVE

Minimalist Tracker for Asynchronous Collaboration

A web-based application designed to bridge the gap between casual chat and heavy project management. Engineered for remote partners working in different time zones.

Focus: "Context Preservation" and "Baton Passing" workflow to reduce anxiety in asynchronous collaboration.

Open Workspace


S
SyncSpace
Global View
WORKSPACE
Alex (Research)
Sarah (Design)
+
Workspace
Projects
Activity
Dark Mode
All Together
Your Turn
Projects waiting for your input
Sarah (Design)
1 Pending
Portfolio Redesign
Collaborating
Ongoing shared projects (Free flow)
No active shared projects.
Waiting
Projects where the ball is in their court

Contact Protocol

// Initialize handshake. Open frequencies.

LINKEDIN
linkedin.com/in/galib-raid

GITHUB
github.com/Galib2003

INSTAGRAM
@riuji_uzu

EMAIL
gar5128@mavs.uta.edu

```
root@portfolio:~$ echo "Based in Arlington, TX"
root@portfolio:~$ echo "Open to opportunities in Embedded Systems & AI"
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